Nguyen Hoang Khang

AI Engineer

khaangnguyeen@gmail.com 0387133745 Vietnam https://github.com/khengyun

https://orcid.org/0000-0003-3616-0367 https://scholar.google.com/citations?user=YpOO60MAAAAJ

linkedin.com/in/khengyun

Combined M.S.–Ph.D. student at Sungkyunkwan University researching intelligent surveillance systems and Edge AI for edge devices. AI researcher and engineer with experience in computer vision, multimodal learning, autonomous systems, medical imaging, and LLM applications, focused on efficient, reproducible AI.

EDUCATION

Combined M.S.–Ph.D. Program 01/2026 – Present
Sungkyunkwan University (SKKU)
Major: Electrical and Electronics Engineering
Scholarship: Fully Funded Scholarship (Full Tuition Waiver and Monthly Stipend)
Research Interests: Intelligent Surveillance Systems, Edge AI, and Edge Devices.

Software Engineering 09/2021 – 04/2025
FPT University
Sub-major: Artificial Intelligence
Cumulative GPA: 3.25/4.0
Research Interests: Computer Vision, Multimodal Learning, Low-level Vision, and Vision-Language Models.

EXPERIENCE

Senior AI Engineer 09/2025 – 02/2026
FPT Software, Vietnam
- Designed and implemented LLM-powered AI Agents to automate core software engineering workflows such as code refactoring, documentation, and bug triage.
- Integrated foundation models (GPT, Claude, LLaMA, etc.) into multi-agent reasoning frameworks (LangChain, AutoGen, CrewAI), enabling complex collaborative reasoning and automated tool orchestration.
- Applied multi-agent design principles to improve productivity, reproducibility, and scalability in enterprise AI development.

AI Researcher 04/2025 – 09/2025
InnoTech Corporation, Vietnam
- Led a cross-functional AI research team in developing and deploying deep learning systems across computer vision, NLP, and recommender systems.
- Optimized research pipelines for high-performance model training, ensuring reproducibility and scalability in production environments.
- Coordinated interdisciplinary collaboration to align AI initiatives with real-world product goals and business requirements.

AI Engineer 04/2024 – 04/2025
Inspire Lab Technology, Vietnam
- Designed and trained deep learning models for medical image analysis and disease classification.
- Built a real-user engagement-based recommendation system for film content personalization.
- Engineered an automated image generation and optimization pipeline to enhance SEO and digital marketing efficiency.

Software Developer 12/2023 – 05/2025
Share A Cart, United States
- Designed and implemented full-stack features with an emphasis on performance optimization and system reliability.
- Collaborated with an international development team to enhance usability and backend efficiency.
- Supported 70% of all integrated retailers through modular system design and API integration.

AI Researcher 03/2022 – 04/2025
FPT University, Vietnam
- Published research on traffic-light detection, lane keeping, and LiDAR-based pedestrian detection.
- Developed and fine-tuned deep learning models for real-world autonomous systems.

PUBLICATIONS

Intuitive Surgical SurgToolLoc and SurgVU Challenges Results: 2022–2025 2026
arXiv:2305.07152 [cs.CV]
Co-author: Khang Hoang Nguyen (InspireLab Technology)

Domain-Specific Image Captioning: Vietnamese Cuisine on the 30VNFoods Dataset 2025
Proceedings of the 2025 10th International Conference on Multimedia Systems and Signal Processing
Huynh Vu Nhu Nguyen, **Khang Nguyen Hoang**, Khac Huy Hynh, Hoang Ngoc Tran

Automated Food Image Labeling for E-Commerce Websites: Combining Content-Based Image Retrieval and Majority Labeling 2025
Proceedings of the 2025 10th International Conference on Multimedia Systems and Signal Processing
Khang Nguyen Hoang, Huynh Vu Nhu Nguyen, Hoang Ngoc Tran

Content-Based Food Image Retrieval Using Pre-Trained Neural Network Models 2025
Proceedings of the 2025 10th International Conference on Intelligent Information Technology
Huynh Nhu Nguyen Vu, **Khang Hoang Nguyen**, Anh Quoc Nguyen, Thanh Tien Hua, Huy Khac Huynh, Hung Huu Nguyen, Hoang Ngoc Tran

Enhancing indoor robot pedestrian detection using improved PIXOR backbone and Gaussian heatmap regression in 3D LiDAR point clouds 2024
IEEE Access
Duy Anh Nguyen, **Khang Nguyen Hoang**, Nguyen Trung Nguyen, Hoang Ngoc Tran

Traffic lights detection and recognition method using deep learning with improved YOLOv5 for autonomous vehicle in ROS2 2023
Proceedings of the 2023 8th International Conference on Intelligent Information Technology
Huy Khanh Hua, **Khang Hoang Nguyen**, Luyl-Da Quach, Hoang Tran Ngoc

Lane road segmentation based on improved UNet architecture for autonomous driving 2023
International Journal of Advanced Computer Science and Applications
Hoang Tran Ngoc, Huynh Vu Nhu Nguyen, **Khang Hoang Nguyen**, Luyl-Da Quach

Optimizing YOLO performance for traffic light detection and end-to-end steering control for autonomous vehicles in Gazebo-ROS2 2023
International Journal of Advanced Computer Science and Applications
Hoang Ngoc Tran, **Khang Hoang Nguyen**, Huy Khanh Hua, Huynh Vu Nhu Nguyen, Luyl-Da Quach

An improved lane-keeping controller for autonomous vehicles leveraging an integrated CNN-LSTM approach 2023
International Journal of Advanced Computer Science and Applications
Hoang Tran Ngoc, Phuc Phan Hong, Nghi Nguyen Vinh, Nguyen Nguyen Trung, **Khang Hoang Nguyen**, Luyl-Da Quach

Combining Autoencoder and Yolov6 Model for Classification and Disease Detection in Chickens 2023
Proceedings of the 2023 8th International Conference on Intelligent Information Technology
Khang Hoang Nguyen, Huynh Vu Nhu Nguyen, Hoang Ngoc Tran, Luyl-Da Quach

Combining Contrast Limited Adaptive Histogram Equalization and Canny's Algorithm for the Problem of Counting Seeds on Rice 2022
Proceedings of the 2022 International Conference on Information Systems for Business Management
Luyl-Da Quach, Phuc Nguyen Trong, **Khang Nguyen Hoang**, Ngon Nguyen Chi

AWARDS

First Prize – Vietnam Student AI Olympiad 2025 (Southern Regional Round) 12/2025
Vietnam Association for Information Processing (VAIP)
Awarded as a member of team FPTU HCM VOAI2.

Third Prize – Vietnam Student AI Olympiad 2025 (National Final) 12/2025
Vietnam Association for Information Processing (VAIP)
Awarded as a member of team FPTU HCM VOAI2.

Top 1 at FleziPT AI Champions Hack The Future Saigon 48 04/2025
FPT Software
An internal 48-hour AI hackathon organized by FPT Software.
It gathers top engineering teams to develop innovative AI solutions addressing real-world challenges. Our team, FHM-AIMPACT_2, proudly won the "Most Favorite Team" award at this season's competition.

2nd Place Leaderboard - Endoscopic Vision Challenge 09/2024
MICCAI 2024 in Marrakesh, Morocco.
Achieved top-ranking performance in surgical tool detection using deep learning models at MICCAI (A*-ranked, CORE Ranking), one of the world's leading conferences in medical image computing and AI in healthcare.

Best Methodology Report Award - Endoscopic Vision Challenge 09/2024
MICCAI 2024 in Marrakesh, Morocco.
Recognized for innovative research methodology and comprehensive analysis in the Surgical Tool Detection challenge, contributing to advancements in medical image segmentation and classification.

First prize for students atthe final of the green startup creative talent contest in Vietnam 11/2022
TECHFEST Vietnam 2022
Developing eco-friendly products from recycled coffee grounds, contributing to waste reduction and sustainable resource utilization.

Consolation Prize at FPT Edu Digital Race 2023 09/2023
Competed in an autonomous vehicle AI challenge, developing real-time object detection and navigation algorithms.

Top 5 Aura Hackathons 07/2022
FPT Sotfware
The only student team to reach the finals. Developed Open Culture, a blockchain-based NFT platform to globalize Vietnamese cultural heritage while addressing intellectual property challenges. Recognized as the Most Inspiring Team for innovative contributions to cultural preservation.

PROJECTS

Innocody Code Assistant @ Innotech 04/2025 – 08/2025
Lead the AI team in building an intelligent IDE extension that automates code generation, debugging, and deployment. Designed a context-aware chat and autocomplete system leveraging internal codebase knowledge, with multi-LLM support. Integrated the assistant with GitHub, GitLab, Docker, and multiple databases to support full-stack developer workflows.

Contract Legal Chatbot @ Innotech 06/2025 – 08/2025
Developed an AI-driven contract review platform, allowing users to upload and analyze PDF contracts through interactive Q&A. The project made a strong impression on the client during the initial demo, resulting in an immediate contract extension for further development.

NotebookLM-MCP 01/2025 – 07/2025
An open-source Python package implementing the Model Context Protocol (MCP), featuring multi-transport support (STDIO, HTTP, SSE), type-safe FastMCP v2 decorators, dynamic tool registration, async task handling, and seamless integration with NotebookLM for AI-driven code and document workflows.

Stock Price Advisory Chatbot @ Unifimoney 05/2025 – 06/2025
Served as an AI consultant supporting the development of a stock price advisory chatbot for Unifimoney. Provided expertise in designing conversational flows and AI models to help users receive real-time stock price recommendations and investment advice.